Game Theory for Multi-Agent Decision Making Exam, 2025
Model Answers

1(A): For any $x \in \Lambda^{(n)}$, $y \in \Lambda^{(m)}$, p and q satisfying the constraints of (NL),

$$x^T Ay \geq -px^T 1_n = -p \text{ and } x^T By \geq -q1_m^T y = -q.$$

It follows that $J(x, y, p, q) = x^T Ay + x^T By + p + q \geq -p - q + p + q = 0$.

So, to confirm that $(\bar{x}, \bar{y}, \bar{p}, \bar{q})$ solves (NL) we must show that $(\bar{x}, \bar{y}, \bar{p}, \bar{q})$ is 'feasible' and $J((\bar{x}, \bar{y}, \bar{p}, \bar{q})) = 0$. The second condition is automatically satisfied because we have chosen $p = -\bar{x}^T A\bar{y}$ and $q = -\bar{x}^T B\bar{y}$. So, it remains to check that $(\bar{x}, \bar{y})$ satisfies the constraints of (NL).

From the Nash equilibrium condition we have $\bar{x}^T By \geq \bar{x}^T B\bar{y}$ for all $y \in \Lambda^{(m)}$.

Taking $y^T = [1, 0, \ldots, 0], \ldots, [0, \ldots, 0, 1]$ and noting $\bar{q} = -\bar{x}^T B\bar{y}$, we see $(\bar{x}^T B)_1 \geq -\bar{q}, \ldots, (\bar{x}^T B)_m \geq -\bar{q}$. This implies $\bar{x}^T B \geq -\bar{q}\ 1_m^T$.

From the Nash equilibrium condition we have $x^T A\bar{y} \geq \bar{x}^T A\bar{y}$ for all $x \in \Lambda^{(n)}$.

We deduce, as above, that $A\bar{y} \geq -\bar{p}\ 1_n$.

We have shown that $(\bar{x}, \bar{y}, \bar{p}, \bar{q})$ is feasible. so $(\bar{x}, \bar{y}, \bar{p}, \bar{q})$ solves (NL).

The cost function of (NL) is convex if $A = -B$, since the nonconvex term in $J(x, p, q)$ then is zero. It is frequently simple to compute solutions to optimization problems when they are convex, because then global minima are local minimal and feasible elements satisfying non-degenerate forms of the Lagrange multiplier conditions are minimizers.

1(B): Player 1 chooses a mixed strategy $(1 - \lambda, \lambda)$. Depending on whether player 2 chooses column $1, 2, \ldots$ or m, the pay-off for player 1, as λ varies from 0 to 1, is

$f_1(\lambda) = a_{11} + \lambda(a_{12} - a_{11})$ (straight line, $f_1(0) = a_{11}$ and $f_1(1) = a_{12}$) .
. .
$f_m(\lambda) = a_{m1} + \lambda(a_{m2} - a_{m1})$ (straight line, $f_m(0) = a_{m1}$ and $f_m(1) = a_{m2}$).

It follows that $(\bar{\lambda}, 1 - \bar{\lambda})$ will be a safety strategy for player 1 if $\bar{\lambda}$ minimizes the upper envelope function $f(\lambda) := \max\{f_1(\lambda), \ldots, f_m(\lambda)\}$.

From the plots of $f_1, \ldots, f_4$ we see the minimizing value of λ occurs at the point $\bar{\lambda}$ where the graphs of f_1 and f_2 intersect, Hence

$$0 + \bar{\lambda} = 1 - \bar{\lambda} \implies \bar{\lambda} = 1/2\,.$$

It follows that the safety strategy for player 1 is $x = (1/2, 1/2)$.

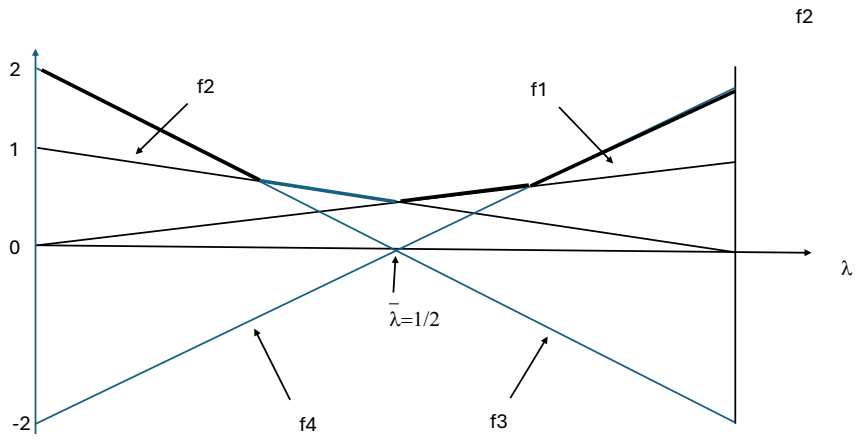


Figure 1

2: Denote

$$p_0 := B \text{ pays}, \; p_1 := \text{B does not pay}$$

and

$$(p_0LL) = B \text{ chooses } p_0 \text{ and } \begin{cases} L \text{ if } A \text{ chooses } L \\ L \text{ if } A \text{ chooses } R \end{cases}$$

$$(p_0LR) = B \text{ chooses } p_0 \text{ and } \begin{cases} L \text{ if } A \text{ chooses } L \\ R \text{ if } A \text{ chooses } R \end{cases}$$

$$\cdots$$

$$\cdots$$

$$(p_0RR) = B \text{ chooses } p_0 \text{ and } \begin{cases} R \text{ if } A \text{ chooses } L \\ R \text{ if } A \text{ chooses } R \end{cases}$$

$$(p_1L) = B \text{ chooses } p_1 \text{ and } L$$

$$(p_1R) = B \text{ chooses } p_1 \text{ and } R\,.$$

From the given pay-off trees we calculate

$A\backslash B$	(p_0LL)	(p_0LR)	(p_0RL)	(p_0RR)	(p_1L)	(p_1R)
L	$(2,-2+a)$	$(2,-2+a)$	$(1,-1+a)$	$(1,-1+a)$	(2,-2)	(1,-1)
R	$(1,-1+a)$	$(1,-1+a)$	$(1,-1+a)$	$(0,a)$	$\boxed{(1,-1)}$	$(0,0)$

.

We see that there is one Nash equilibrium (R, p_1L). In words,

'A chooses R' and 'B does not pay and chooses L'.

Neither the row nor the column containing this Nash equilibrium is dominated. It follows that the Nash equilibrium is admissible.

For this example, B obtains no advantage from choosing to buy information.

3(A). (i): The profits for the 3 companies are

$$L^A(a,b,c) = a(12-a-b-c) - 2a$$
$$L^B(a,b,c) = b(12-a-b-c) - 4b$$
$$L^C(a,b,c) = c(12-a-b-c) - 4c$$

(ii): Setting $\partial/\partial a(..) = \partial/\partial b(..) = \partial/\partial c(..)$ to get Nash equilibrium $(\bar{a},\bar{b},\bar{c})$ and assuming $\bar{b} = \bar{c}$ gives:

$$12 - \bar{a} - 2\bar{b} - \bar{a} = 2 \implies \bar{a} + \bar{b} = 5$$
$$12 - \bar{a} - 3\bar{b} - \bar{b} = 4 \implies \bar{a} + 3\bar{b} = 8$$

These have solution $\bar{a} = 7/2$ and $\bar{b} = \bar{c} = 3/2$.

(iii): Now assume A is the leader. If A chooses a, B and C will choose production levels $b(a) = c(a)$ to satisfy $\partial/\partial b\, L^B(a,b,c) = 0$. Hence

$$12 - a - 2b(a) - c(a) = 12 - a - 3b(a) = 4 \implies b(a) = c(a) = (8-a)/3.$$

The pay-off for A is then $a(12 - a - (2/3)(8-a)) - 2a$. a^* is obtained by setting $\partial/\partial a\,(..) = 0$, giving

$$\partial/\partial a \left(a^*(\frac{20}{3} - \frac{1}{3}a^*) - 2a^* \right) = \frac{20}{3} - \frac{2}{3}a^* - 2 \implies a^* = 7.$$

The the production levels of B and C are $b^* = c^* = (8-a^*)/3 = 1/3$.

(iv): If $(\bar{a},\bar{b},\bar{c})$ lies in the responses surfaces, we have

$$\bar{a} = a(\bar{b},\bar{c}), \bar{b} = b(\bar{a},\bar{c}) \text{ and } \bar{c} = c(\bar{a},\bar{b})\,.$$

Then, by definition of the response surfaces,

$$\bar{a} = \arg\min L^A(a,\bar{b},\bar{c}), \bar{b} = \arg\min L^B(\bar{a},b,\bar{c}), \bar{c} = \arg\min L^C(\bar{a},\bar{b},c)$$

whence

$$L^A(a,\bar{b},\bar{c}) \geq L^A(\bar{a},\bar{b},\bar{c}), L^B(\bar{a},b,\bar{c}) \geq L^B(\bar{a},\bar{b},\bar{c}), L^C(\bar{a},\bar{b},c) \geq L^C(\bar{a},\bar{b},\bar{c})$$

for any production levels (a,b,c). This means $(\bar{a},\bar{b},\bar{c})$ is a Nash equilibrium.

4(A). (i): Write $a(x) = [a_1(x), a_2(x)]^T$. '$\bar{x}$ is an evolutionary Nash equilibrium' means

$$x^T a(\bar{x}) \leq \bar{x}^T a(\bar{x}), \text{ for all } x \in \Lambda^{(2)}.$$

$\bar{x}$ is 'stable' if there exists $\epsilon > 0$ such that

$$x^T a(x) < \bar{x}^T a(x), \text{ for all } x \in \Lambda^{(2)} \text{ such that } ||x - \bar{x}|| \leq \epsilon.$$

(ii) Any $x \in \Lambda^{(2)}$, $x \neq \bar{x}$, that is 'close' to $\bar{x}$ can be written $x = (\bar{x}_1 + \epsilon, \bar{x}_2 - \epsilon)$ for all $\epsilon \neq 0$, sufficiently small.
Suppose first $\epsilon > 0$. Then, since $(a_1(x) - a_2(x)) - (a_1(\bar{x}) - a_2(\bar{x})) < 0$,

$$x^T a(x) - \bar{x}^T a(x) = \epsilon(a_1(x) - a_2(x)) < \epsilon(a_1(\bar{x}) - a_2(\bar{x}))$$

But since $\bar{x}$ is a Nash equilibrium

$$\epsilon^{-1}(x^T - \bar{x}^T)a(\bar{x}) = (a_1(\bar{x}) - a_2(\bar{x})) \leq 0\,.$$

It follows that $x^T a(x) - \bar{x}^T a(x) < 0$.
If $\epsilon < 0$, the same analysis, but now using the condition $(a_1(x) - a_2(x)) - (a_1(\bar{x}) - a_2(\bar{x})) > 0$ gives, once again, $x^T a(x) - \bar{x}^T a(x) < 0$. Tbis means $\bar{x}$ is a stable equilibrium.

(iii): the replicator equations are

$$\begin{cases} \dot{x}_1 = x_1\Big(a_1(x) - (x_1 a_1(x_1) + x_2 a_2(x))\Big) \\ \\ \dot{x}_2 = x_2\Big(a_2(x) - (x_1 a_1(x_1) + x_2 a_2(x))\Big) \end{cases}$$

If the interior Nash equilibrium $\bar{x}$ is stable then, for any initial state $x(0)$ close to $\bar{x}$, we expect the solution $x(t) \longrightarrow \bar{x}$ as $t \longrightarrow \infty$. If $\bar{x}$ is not stable we expect there will be initial conditions arbitrary close to $\bar{x}$ such that $x(t) \not\longrightarrow \bar{x}$ as $t \longrightarrow \infty$.

4(B): Suppose $a_1(x) = -2x_1 - 2x_2^2$ and $a_2(x) = -3x_2$. Then $(\bar{x}_1, \bar{x}_2)$ is a Nash equilibrium when

$$a_1(\bar{x}_1, \bar{x}_2) = a_2(\bar{x}_1, \bar{x}_2) \text{ and } (\bar{x}_1, \bar{x}_2) \in \Lambda^{(2)}.$$

Writing $\bar{x}_1 = (1 - \bar{x}_2)$, we require $-2(1 - \bar{x}_2) - 2\bar{x}_2^2 = -3\bar{x}_2$. This has one strictly interior solution $\bar{x}_2 = \frac{1}{2}$. So the strictly interior Nash equlilibrium is $(\bar{x}_1, \bar{x}_2) = (\frac{1}{2}, \frac{1}{2})$.

Take any $\epsilon \neq 0$. Then, for $(\bar{x}_1, \bar{x}_2) = (\frac{1}{2}, \frac{1}{2})$, we have

$$\begin{aligned} a_1(\bar{x}_1 + \epsilon, \bar{x}_2 - \epsilon) - a_2(\bar{x}_1 + \epsilon, \bar{x}_2 - \epsilon) - (a_1(\bar{x}_1, \bar{x}_2) - a_2(\bar{x}_1, \bar{x}_2)) \\ = -\epsilon^2 - 3\epsilon\,. \end{aligned}$$

Notice that, for small $\epsilon \neq 0$, this expression is negative if $\epsilon > 0$ and positive if $\epsilon < 0$. So the Nash equilibrium is stable, by Part (A).

5(A): For the given problem data, the Bellman equation is

$$\begin{cases} V_{t-1} = \min\{V_t(x+u) + u^2\} \text{ for } t = 1, \ldots, N \\ V_N(x) = x^2 \,. \end{cases}$$

Assume $V_t(x) = p_t x^2$, for $t = 0, \ldots, N$, for some sequence $p_0, \ldots, p_N$. Then, from the Bellman equation we have, for any t,

$$p_{t-1}x^2 = \min\{(p_t(x+u)^2 + u^2 : u\} = p_t x^2 + \min\{2p_t xu + (p_t+1)u^2 : u\} = p_t x^2 - \frac{p_t^2}{p_t+1}x^2$$

and the minimizing u is $u = -\frac{p_t}{p_t+1}\, x$.

Since, for each t, the Bellman equation must be satisfied for all x, $\{p_t\}$ satisfies the recursion

$$p_{t-1} = p_t - \frac{p_t^2}{p_t+1} = \frac{p_t}{p_t+1}, \text{ for } t = 1. \ldots, N. \tag{0.1}$$

These equations can be solved by backwards recursion, with right boundary condition $p_N = 1$, from the dynamic programming condition $V_N(x) = x^2$, for all x.

From the theory of dynamic programming, The minimizing control is given by the feedback relation

$$u_{t-1} = \frac{p_t}{p_t+1} x_{t-1}, \text{for } t = 1, \ldots, N.$$

and the minimum cost is $p_0 x_0^2$.

5(B): We know from part (A) that, for a given x_0, the follower's minimum cost is

$$\min\{J(x_0, \{u_0, \ldots, u_{N-1}\}) : \{u_0, \ldots, u_{N-1}\}\} = p_0 x_0^2.$$

It follows that the leader's profit is

$$p_0 x_0^2 - g(x_0) = p_0 x_0^2 - 3x_0^3.$$

Notice that $p_0 > 0$, by (0.1) and since $p_N = 1$.

The Steckelberg minimizer $\bar{x}_0$ is then given by

$$d/dx_0 \; (p_0 \bar{x}_0^2 - g(\bar{x}_0)) = 2p_0 \bar{x}_0 - 9\bar{x}_0^2 \,.$$

II follows that the Stackelberg optimizer is

$$\bar{x}_0 = \frac{2}{9} \times p_0,$$

in which p_0 is given by the recursion (0.1). (Notice we can ignore root $\bar{x}_0 = 0$ since this gives only a local minimum.)

The follower strategy $\{\bar{u}_0, \ldots, \bar{u}_{N-1}\}$ is the optimal control found in part A, for initial state $\bar{x}_0$.